Day 2 — Assignment

Pandas Practice & Visualization — winequality-white.csv

Work through these tasks using only what we covered today. Target time: 30–45 minutes. Use winequality-white.csv throughout — remember it's semicolon-separated.

Section 1 — Pandas Practice

Task 1: Import pandas, read winequality-white.csv into a DataFrame, and run shape and info() to confirm it loaded correctly.

Task 2: Display the first 7 values of the 'volatile acidity' column.

Task 3: Build a mask for rows where volatile acidity is less than 0.3, and count how many rows match.

Task 4: Add a second condition to that same mask — quality of 7 or higher — and count how many rows match both conditions together.

Task 5: Group the data by quality and find the average pH for each quality score.

Task 6: Sort the DataFrame by residual sugar in descending order and show the top 8 rows, displaying only the residual sugar and quality columns.

Section 2 — Visualization with pandas' built-in .plot()

Task 7: Create a histogram of the pH column.

Task 8: Create a scatter plot with sulphates on the x-axis and alcohol on the y-axis.

Task 9: Create a box plot of every column in the DataFrame at once. Then create a second box plot using only fixed acidity and alcohol. What's different about how readable the two box plots are, and why?

Task 10: Create a box plot of the citric acid column on its own.